

NNUE vs MCE Rank Correlation Report

Data: overnight/rank_corr_raw_50g_100r.jsonl
Games: 50 Positions: 1000 Candidates/turn: ~90
Rollouts per candidate: 100

--- Game Scores (player 0) ---

Base: mean=94.0 std=2.9 min=83 max=101
bear: mean= 23.4 std= 6.6 min= 4 max= 27
elk: mean= 8.1 std= 4.2 min= 2 max= 17
salmon: mean= 8.5 std= 6.3 min= 0 max= 25
hawk: mean= 8.9 std= 5.0 min= 2 max= 26
fox: mean= 13.2 std= 3.9 min= 3 max= 22
hab: 29.0 wl_total: 62.0 tokens: 2.9

--- Rank Correlation ---

Kendall tau: 0.484 (std=0.208)
Spearman rho: 0.628 (std=0.243)
Top-1 agree: 22.9% (229/1000)

--- MCE Top-1 in NNUE Top-X ---

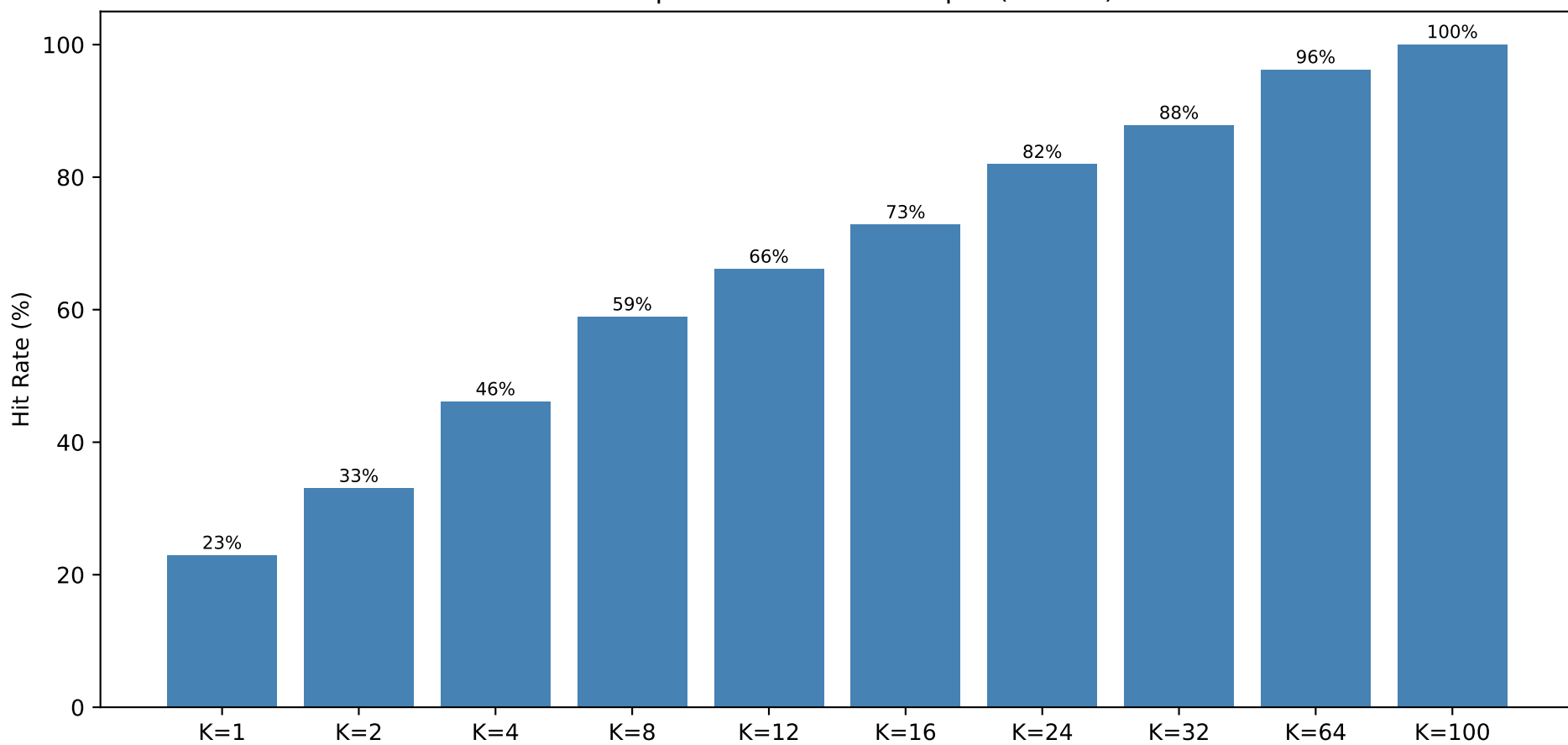
K= 1:	22.9% hit	77.1% miss
K= 2:	33.0% hit	67.0% miss
K= 4:	46.1% hit	53.9% miss
K= 8:	58.9% hit	41.1% miss
K= 12:	66.1% hit	33.9% miss
K= 16:	72.8% hit	27.2% miss
K= 24:	81.9% hit	18.1% miss
K= 32:	87.8% hit	12.2% miss
K= 64:	96.2% hit	3.8% miss
K=100:	100.0% hit	0.0% miss

Avg gap when K=8 misses: 2.01 pts

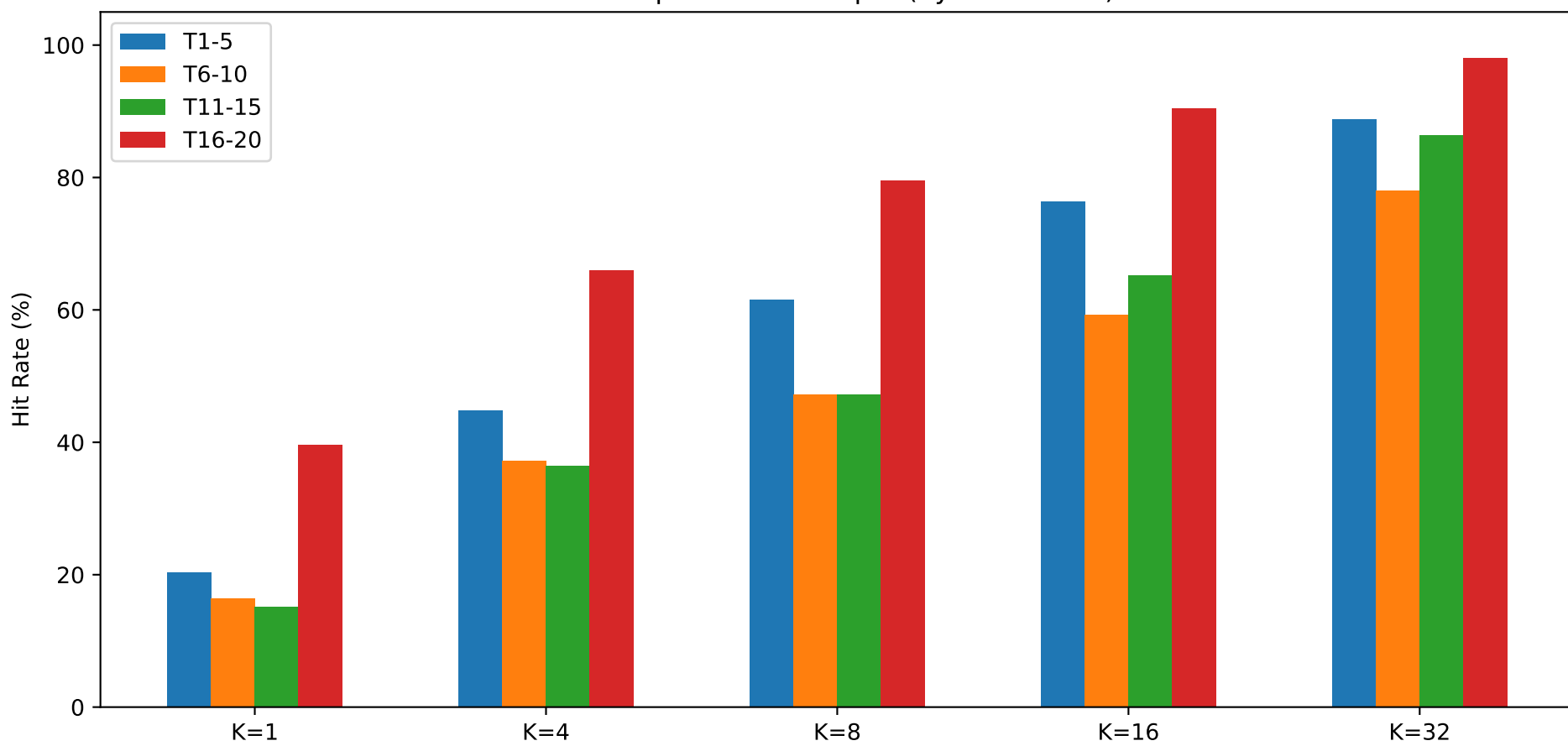
--- MCE Rollout Std Dev ---

All cands: mean=2.70
MCE top-1: mean=2.62
NNUE top-1: mean=2.68

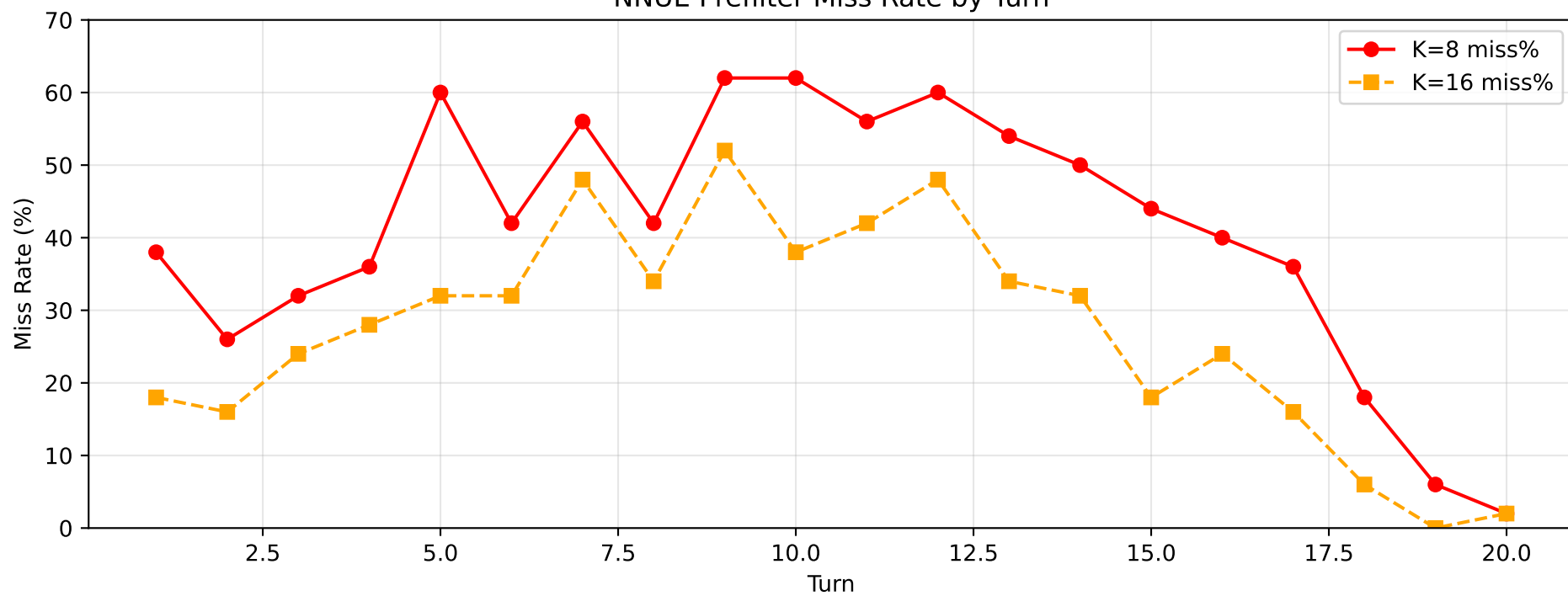
MCE Top-1 Found in NNUE Top-K (Overall)



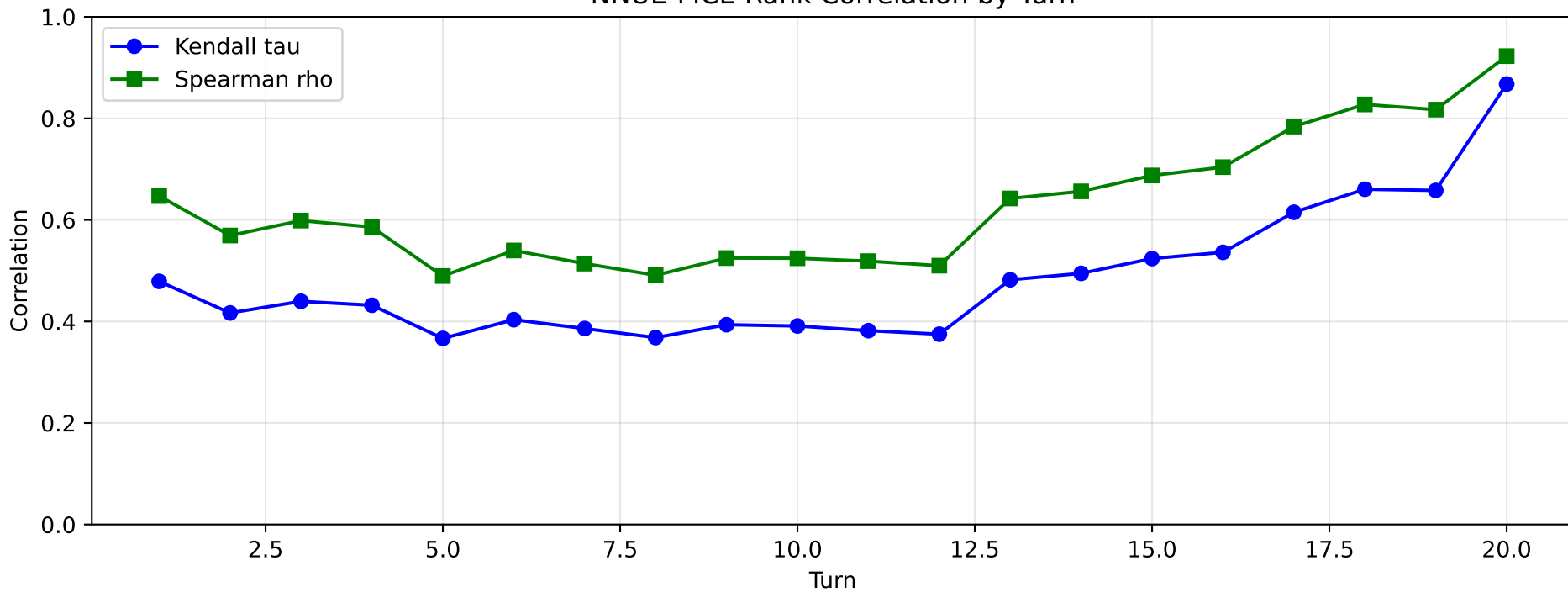
MCE Top-1 in NNUE Top-K (by Turn Phase)



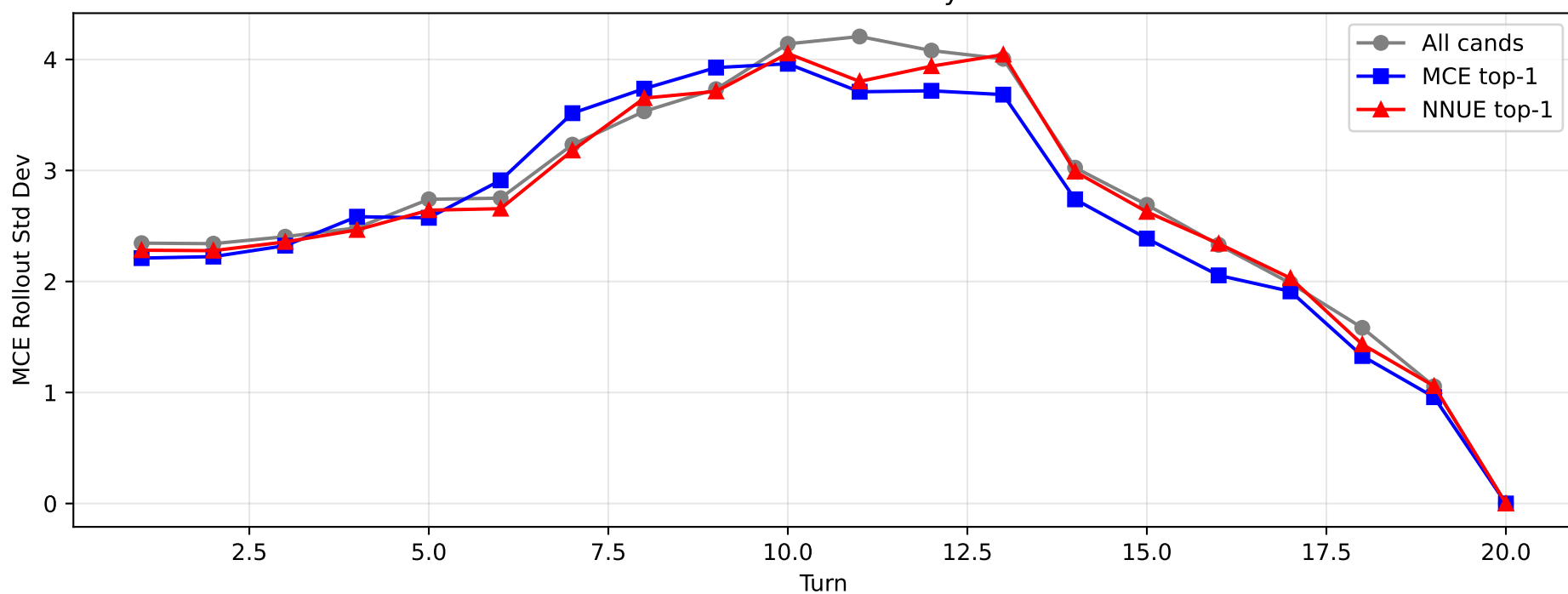
NNUE Prefilter Miss Rate by Turn



NNUE-MCE Rank Correlation by Turn

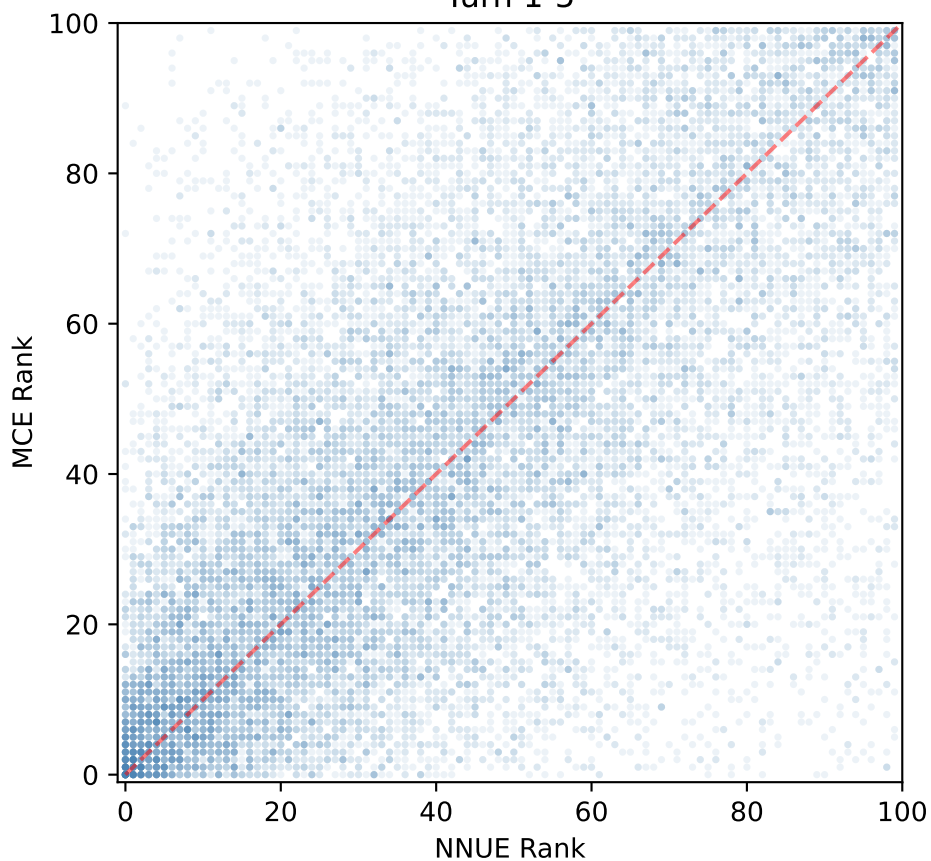


MCE Evaluation Noise by Turn

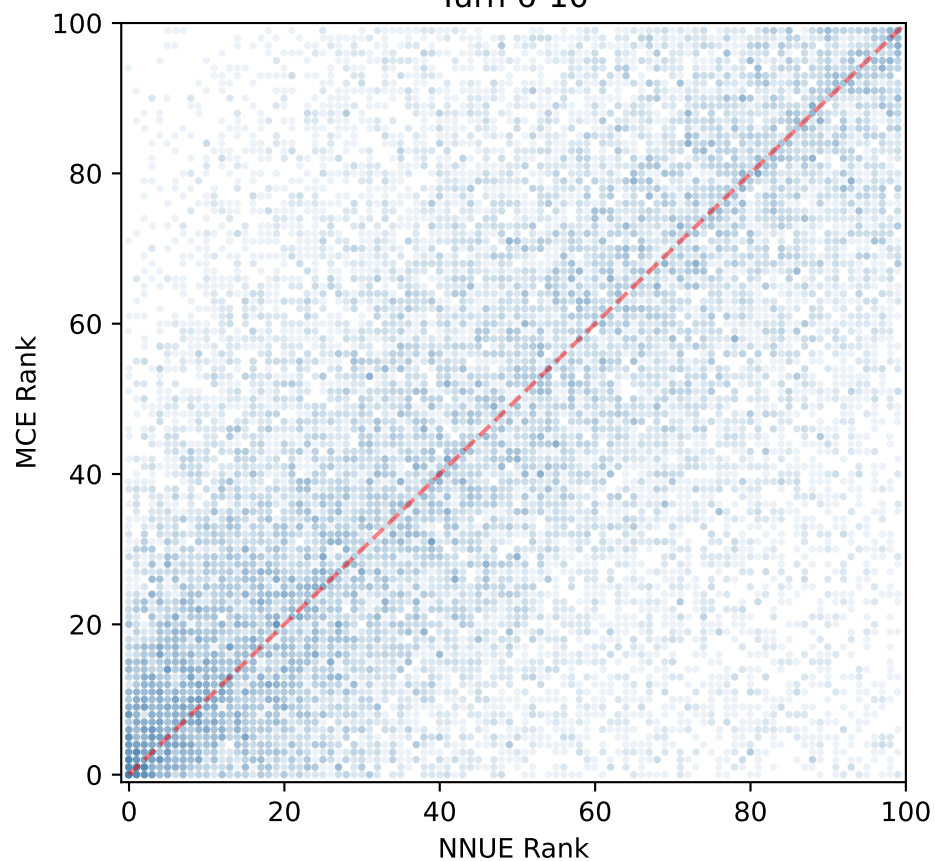


NNUE Rank vs MCE Rank (each dot = one candidate)

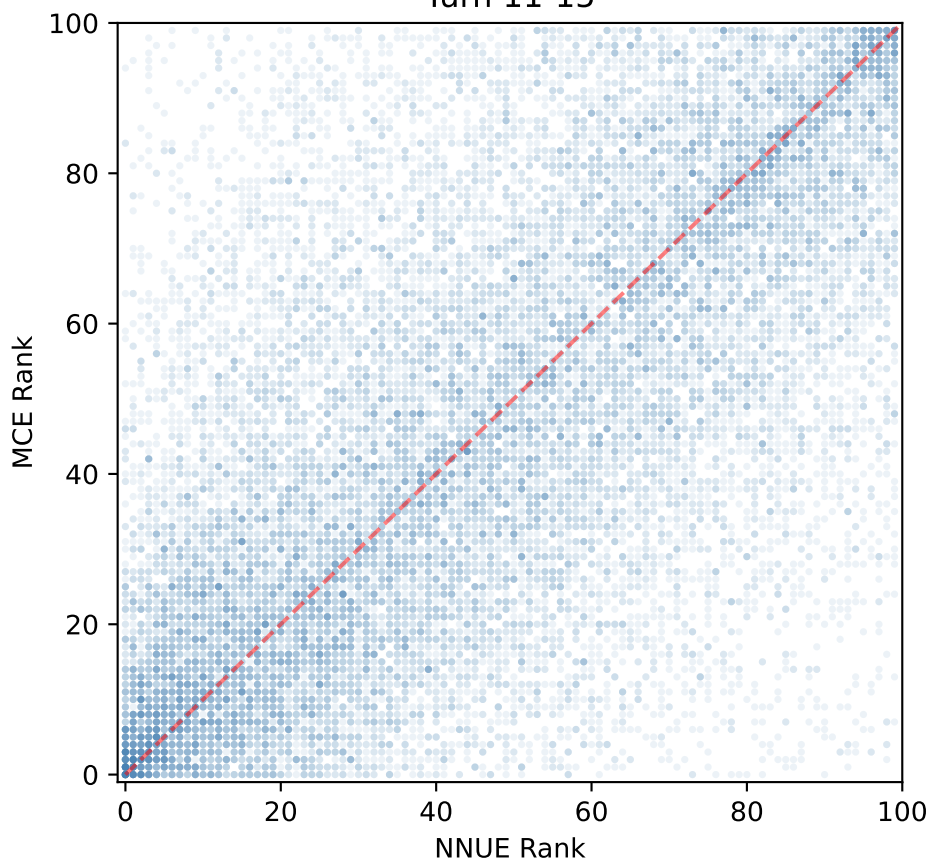
Turn 1-5



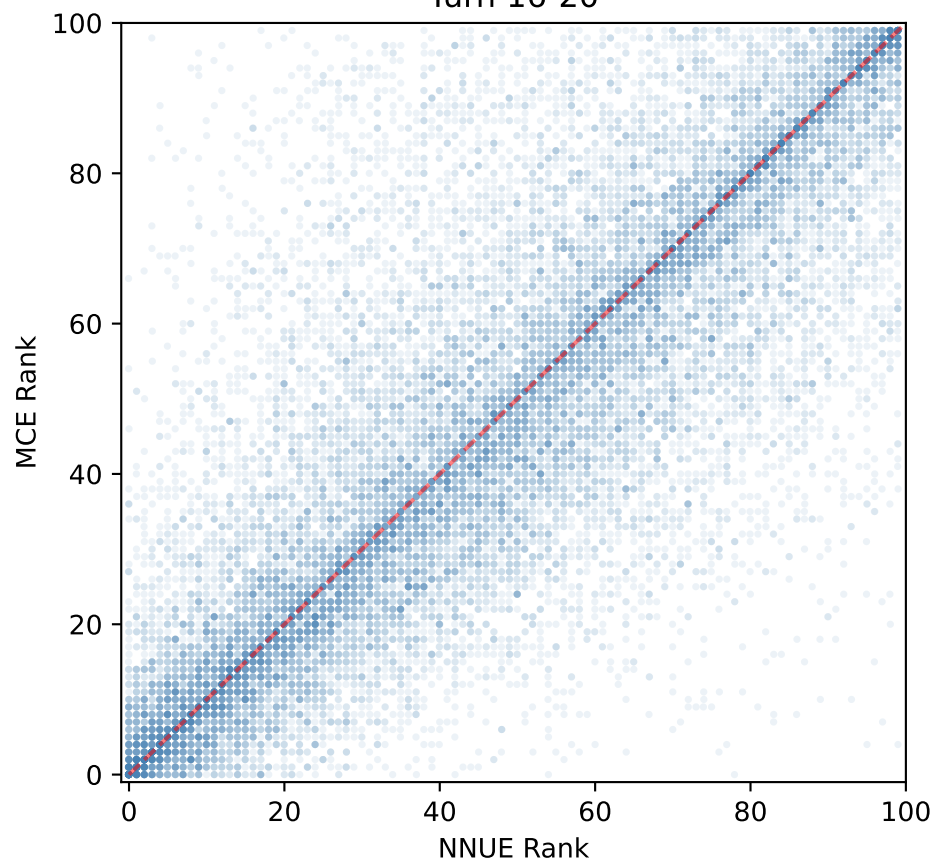
Turn 6-10



Turn 11-15



Turn 16-20



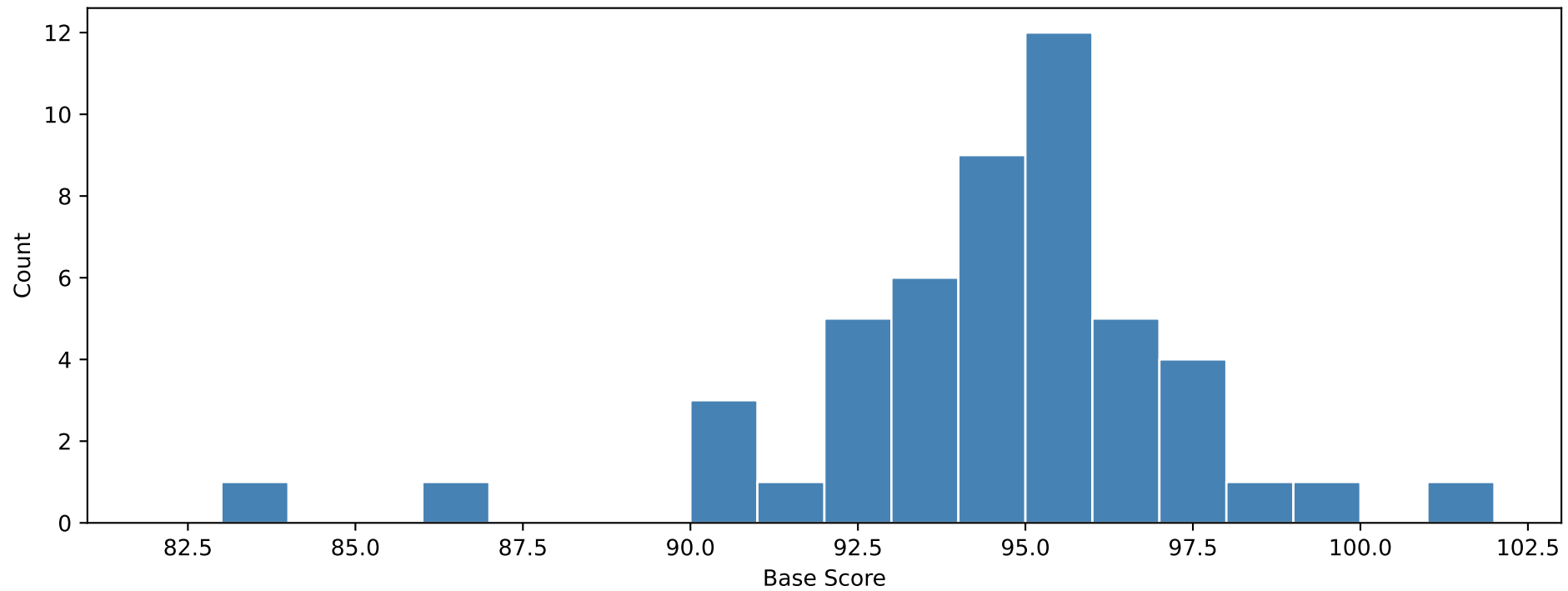
Per-Turn Detail Table

Turn	tau	rho	miss8	miss16	gap	std	agree	n
1	0.479	0.647	38%	18%	0.9	2.35	16%	50
2	0.417	0.569	26%	16%	0.7	2.34	28%	50
3	0.440	0.599	32%	24%	0.8	2.40	20%	50
4	0.432	0.586	36%	28%	1.7	2.48	24%	50
5	0.367	0.489	60%	32%	1.4	2.74	14%	50
6	0.403	0.540	42%	32%	2.6	2.75	18%	50
7	0.386	0.514	56%	48%	2.9	3.23	18%	50
8	0.368	0.491	42%	34%	3.4	3.53	22%	50
9	0.394	0.525	62%	52%	2.6	3.73	12%	50
10	0.391	0.524	62%	38%	3.3	4.14	12%	50
11	0.382	0.519	56%	42%	2.5	4.21	4%	50
12	0.375	0.510	60%	48%	2.8	4.08	14%	50
13	0.482	0.642	54%	34%	2.6	4.01	18%	50
14	0.495	0.656	50%	32%	1.1	3.03	16%	50
15	0.524	0.687	44%	18%	1.0	2.69	24%	50
16	0.536	0.704	40%	24%	0.8	2.33	20%	50
17	0.615	0.784	36%	16%	1.2	1.98	24%	50
18	0.660	0.828	18%	6%	0.7	1.58	32%	50
19	0.658	0.817	6%	0%	0.2	1.06	42%	50
20	0.868	0.923	2%	2%	1.0	0.00	80%	50

Top 20 Disagreements (NNUE top-1 != MCE top-1, by MCE gap)

G10	T10:	NNUE→mkt1(MCE rk 14)	MCE→mkt1(NNUE rk 14)	gap=13.9
G19	T 7:	NNUE→mkt3(MCE rk 17)	MCE→mkt3(NNUE rk 91)	gap=9.5
G44	T13:	NNUE→mkt3(MCE rk 45)	MCE→mkt0(NNUE rk 21)	gap=9.3
G20	T12:	NNUE→mkt0(MCE rk 47)	MCE→mkt3(NNUE rk 34)	gap=9.0
G 0	T10:	NNUE→mkt2(MCE rk 38)	MCE→mkt2(NNUE rk 34)	gap=8.9
G 1	T 7:	NNUE→mkt1(MCE rk 54)	MCE→mkt1(NNUE rk 15)	gap=8.8
G44	T10:	NNUE→mkt0(MCE rk 94)	MCE→mkt0(NNUE rk 11)	gap=8.5
G43	T12:	NNUE→mkt0(MCE rk 17)	MCE→mkt1(NNUE rk 25)	gap=8.5
G20	T13:	NNUE→mkt0(MCE rk 84)	MCE→mkt0(NNUE rk 25)	gap=8.4
G31	T 8:	NNUE→mkt0(MCE rk 34)	MCE→mkt2(NNUE rk 69)	gap=8.2
G 6	T 5:	NNUE→mkt3(MCE rk 32)	MCE→mkt0(NNUE rk 38)	gap=8.1
G36	T12:	NNUE→mkt3(MCE rk 25)	MCE→mkt3(NNUE rk 19)	gap=7.8
G 1	T 8:	NNUE→mkt1(MCE rk 6)	MCE→mkt2(NNUE rk 92)	gap=7.8
G48	T 6:	NNUE→mkt0(MCE rk 16)	MCE→mkt2(NNUE rk 77)	gap=7.8
G23	T 8:	NNUE→mkt0(MCE rk 20)	MCE→mkt2(NNUE rk 17)	gap=7.7
G10	T 4:	NNUE→mkt1(MCE rk 14)	MCE→mkt1(NNUE rk 17)	gap=7.5
G32	T 9:	NNUE→mkt1(MCE rk 15)	MCE→mkt1(NNUE rk 28)	gap=7.1
G47	T 6:	NNUE→mkt3(MCE rk 22)	MCE→mkt1(NNUE rk 41)	gap=6.9
G 0	T11:	NNUE→mkt3(MCE rk 11)	MCE→mkt2(NNUE rk 90)	gap=6.7
G36	T10:	NNUE→mkt1(MCE rk 11)	MCE→mkt3(NNUE rk 3)	gap=6.7

Game Score Distribution (n=50, mean=94.0)



Wildlife Score Distribution

